

Robot Trajectory Optimization

Teaching a robot to steer itself along a path — safely, using math instead of guesswork.



What it does

Works out the exact steering angle and speed a robot needs, instant by instant, so it follows a path without ever crashing.



How it decides

Solves a small math problem at every step instead of searching a map like a maze-solver



Why it matters

Same idea (Model Predictive Control) used in real self-driving cars and robots, built here from scratch for a university course.

HOW IT WORKS

Sense, Plan, Solve, Act — Ten Times a Second



Sense

Where is the robot right now - position, heading, speed.



Plan

Redraw a smooth target path starting from here.



Solve

An Active-Set QP solver, wrapped in an SQP loop, finds the best steering + acceleration for the next 2 seconds - never breaking the robot's limits.



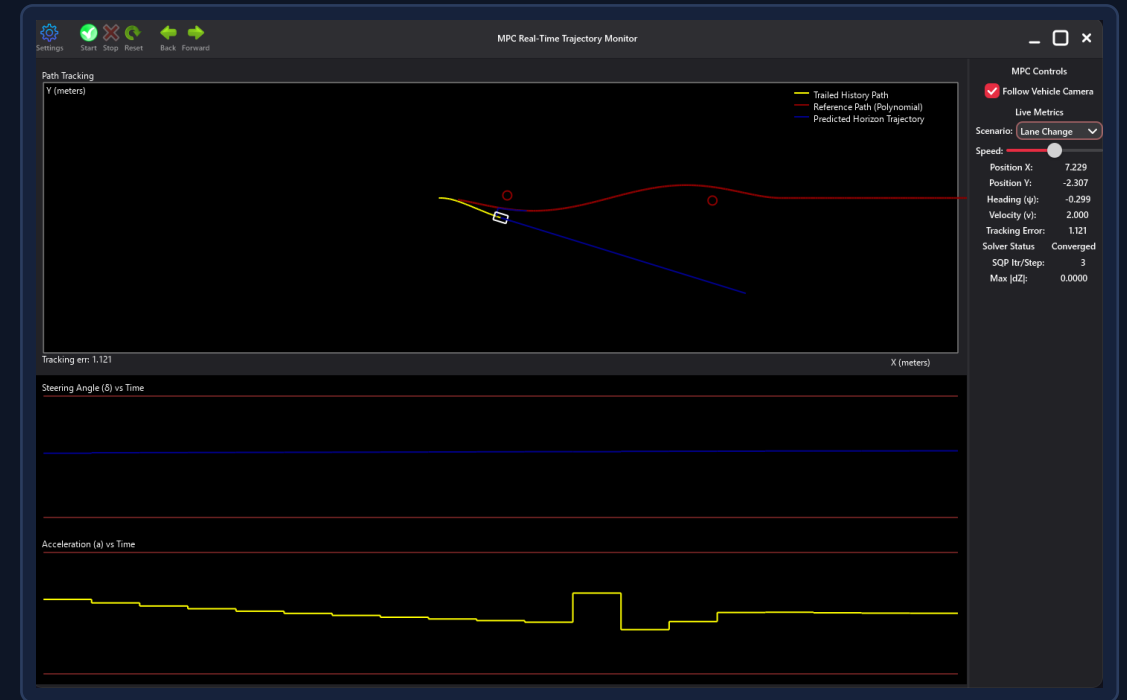
Act

Send just that first move to the real robot.



Repeat

Do it all again, about 10 times a second.



Live run — yellow: driven path · red: target path · blue: direction / calculation offset

BUILT WITH

C++17

natID framework

Sparse LDL^T solver

Active-Set QP + SQP

Try It Yourself

- **Real-time GUI**

Live path, predicted horizon & obstacles, redrawn every step.

- **Tunable solver settings**

Iterations, tolerance and step size, adjustable live.

- **Cross-platform**

Windows, Linux, macOS (Intel & Apple Silicon).

- **3 built-in scenarios**

Straight Line, Lane Change, S-Curve.

- **6 UI languages**

English, Bosnian, German, Spanish, French, Japanese.

- **Open source**

MIT license, C++17, no outside libraries.

DOWNLOAD

Windows ↓

Linux ↓

macOS — Apple Silicon ↓

macOS — Intel ↓

github.com/ehadziabdic/RobotTrajectoryOpt

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